Praneat Data

Third Year Undergraduate | Department of Mathematics

praneat21@iitk.ac.in | +91-7974276176
Praneat Data | praneatdata

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	BS	Indian Institute of Technology, Kanpur	7.74/10
2021	CBSE(XII)	The Shishukunj International School, Indore	97.2%
2019	CBSE(X)	The Shishukunj International School, Indore	94.4%

ACHIEVEMENTS

- Bagged **Bronze Medal in Inter-IIT Tech Meet 11.0** in 'Drona Aviation's Pluto Drone Swarm Challenge'
- Secured **All India Rank 868** in **JEE Advanced 2021** and **All India Rank 2937** in **JEE Mains 2021**
- **All India Rank 912** - Recipient of the **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship** in 2020
- Among the **top 30** students in the state to qualify **Regional Mathematics Olympiad (HBCSE)** in 2018 and 2021
- Qualified the **Indian Olympiad Qualifier in Astronomy**, conducted by **IAPT and HBCSE** in 2019 and 2020.
- Qualified the **Indian Olympiad Qualifier in Chemistry**, conducted by **IAPT and HBCSE** in 2020.

COMPETITIONS

Drona Aviation's Pluto Drone Swarm Challenge - Inter IIT Tech Meet 11.0 *(January'23-February'23)*

Objective	Design and implement a communication wrapper to facilitate drone control for **Pluto 2.0**, integrating a **webcam-based** position detection system to develop an efficient and accurate position control system for the drone
Approach	• Designed a wrapper that can communicate over WiFi through **MSP packets**, by an **ESP module on-board** • Utilized a single **O-cam camera** to design and implement a robust 3D positioning system • Developed algorithms to enable real-time 3D tracking of **Aruco** marker on a drone, allowing for precise tracking • The position control system effectively regulated the roll, yaw, pitch, and throttle (RYPT) of the drone • Employed the communication wrapper and position control system to guide a drone along a designated path
Result	Won the **Bronze Medal** in the High preparation competition amongst the 22 participating IITs

KEY PROJECTS

- **DataScience with R** *(May' 23- July' 23)*
  Mentee under **Stamatics Society**,IITK (Github)
  - Introduced the fundamental concepts of Data Science, enabling the transformation of raw data into meaningful insights.
  - Implemented web scraping, data cleaning, dataset processing, proficient in dataset manipulation, visualization, and plotting.
  - Utilized widely used libraries such as **tidyverse, rvest, dplyr, MASS, txhousing, iris, and ggplot2.**
  - Experienced in data profiling, analysis, and proficient in integrating C++ functions into R workflows.
- **Application of Machine Learning | Course Project** *(January' 23- April' 23)*
  Prof. Purushottam Kar | CS771A - Introduction to Machine Learning (Github)
  - Developed a **linear machine learning model** to crack and analyse the **cryptographic algorithm XORRO**.
  - Implemented a **decision tree learning algorithm** for playing the word-guessing game, on a given dictionary of words.
  - Conducted a comprehensive analysis of linear machine learning models, evaluating their performance based on **MAE values**.
  - Worked in a team of **5 members** with efficient distribution of work and documentation of the work.
- **Project Mentor, Introduction to Geometric Deep Learning | Stamatics Society, IITK** *(May'23-July'23)*
  - Explored foundations of Geometric Deep Learning, inspired by Michael Bronstein and Petar Veličković.
  - Developed a comprehensive understanding of implementing Neural Networks for graphs and networks.
  - Designed **Convolutional Graph Neural Network** for **Music Track Recommendation** in Spotify playlists.
- **Object detection using OpenCV** *(December'22)*
  Shaastra'22 | Aerial Robotics, IITK (Github)
  - Implemented object detection using **OpenCV**, distinguishing objects based on color and the presence of Aruco markers.
  - Calibrated an **NVIDIA Jetson** and a **O-CAM** to calculate the pose of detected Aruco with respect to world frame.
  - Leveraged the capabilities of the developed pose estimator to successfully implement a pick-up and drop mechanism.
- **Trajectory Generation | Aerial Robotics** *(July'22- December'22)*
  - Developed an algorithm to generate smooth and jerk-free trajectories in 3D space, utilizing given waypoints as input.
  - Explored the work of Miguel Fernandez and David Perez on **Local Gaussian Modifiers** to develop dynamic trajectories.
  - Implemented the generated trajectory to navigate the drone autonomously and follow the designated trajectory.

TECHNICAL SKILLS AND RELEVANT COURSES *(*: Ongoing)*

Programming Languages	C, C++, HTML, LaTeX, Python, R
Libraries	OpenCV, Numpy, Scikit-Learn, PyTorch, NetworkX.
Mathematics Courses	Probability and Statistics | Introduction to Linear Algebra | Abstract Algebra | Set Theory and Logic | Time Series and Analysis* | Real Analysis | Complex Analysis*
Computer Science Courses	Fundamental of Computing (A) | Introduction to Machine Learning (A)

POSITIONS OF RESPONSIBILITY

- **Senior Team Member** | Aerial Robotics (SNT Team) *(May'23-Present)*
- **Coordinator** | Photography Club (Media and Cultural Council) *(May'23-Present)*
- **Editor Media** | Vox Populi (Journalism body, IIT Kanpur) *(May'23-Present)*